Akul Singh

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Education

University of California, Santa Barbara Santa Barbara, CA
B.S. in Computer Science – GPA: 3.97; Dean's Honors *Expected Graduation: March 2027*

- **Relevant Coursework:** Artificial Intelligence, Mobile App Development, Object-Oriented Programming

Experience

Toyota Automated Logistics June 2025 – Present
Software Engineering Intern *Boise, ID*

- Implemented mixed precision and quantization-aware training experiments with A/B testing, leading to improved end-to-end accuracy by 4% in automation pipelines while reducing training time on limited GPU hardware.
- Developed Agentic AI workflows leveraging Vision Language Models to auto-tag image datasets for intelligent dataset pruning and labeling, replacing manual annotation pipelines & labeling throughput by 50%.
- Built a Docker-Containerized tool-chain to ingest and manage image datasets, supporting multiple image-analysis features at scale.
- Ran continual learning and fine tuning trainings on robotic vision data, benchmarking replay rates to determine the most resource-efficient update strategies.

CafeAI Startup Feb 2026 – Present
Machine Learning Engineer Intern *Remote*

- Engineered end-to-end data preparation workflows for a vision-based tray inspection system, including dataset curation, annotation, and validation-set processing for robust object detection performance.
- Built synthetic data pipelines for image augmentation and compositing, reducing manual data curation requirements and accelerating model iteration from small customer-provided datasets.
- Trained and benchmarked transformer-based detection model (RF-DETR) for food-item recognition, achieving 96% precision, while also developing hand-detection safeguards to filter occluded frames and improve real-world inference reliability.

Systems and Networking Research Lab, UCSB Sept 2024 – May 2025
Software Engineering Research Assistant *Santa Barbara, CA*

- Designed & implemented NetFlex, a network-quality measurement tool that integrates LLM-powered Retrieval-Augmented Generation to diagnose regional speedtest failures with contextual government datasets.
- Built a FastAPI backend to ingest time-series network measurements and query PostgreSQL database for similar historical network data using geospatial lookups and latency-aware joins.
- Engineered data pipelines to correct for location bias in historical network records, improving LLM-based analysis precision by 25%.

Singapore University of Technology and Design Jun 2024 – Aug 2024
AI Research Intern *Singapore*

- Developed JupyterBot, an AI learning assistant built upon an LLM integrated with JupyterLab to guide students in 3 classes through programming assignments
- Used React.js and Typescript to implement Chatbot Context Management for improved LLM output quality.
- Extended LangChain to optimize AI assistant's logic-based guidance system to adaptively use students' existing code and logic to approach the optimal solution, improving solution relevance and quality by 20%.

Projects

CanvasClash | *Next.js, Tailwind CSS, Express.js, Node.js, Socket.io, TensorFlow.js* Mar 2025

- Led team of 4 to 3rd place in the ACM Winter Project Series with a real-time multiplayer drawing-based AI game.
- Developed a dynamic room generation system using Socket.io, supporting 10+ concurrent multiplayer rooms.
- Integrated TensorFlow.js on the Next.js frontend for AI-powered sketch recognition, achieving up to 90% accuracy within milliseconds.

QoS Trends Prediction Model | *Python, PyTorch, Scikit-learn, Pandas* Jan 2025

- Designed a machine learning pipeline to predict Quality of Speed trends by training a deep learning model on packet data, achieving an accuracy increase of 18% through early stopping and hyperparameter optimization.
- Enhanced data quality by implementing advanced data cleaning and normalization techniques, alongside iterative data collection methods, which contributed to a 15% improvement in model accuracy and reliability.

Technical Skills

Languages: Python, C++, SQL, JavaScript, TypeScript, Java, HTML/CSS, MIPS Assembly
Frameworks & Libraries: Linux, MongoDB, Express.js, FastAPI, React.js, Node.js, LangChain, Angular, PyTorch
Developer Tools: Git, Docker, Azure DevOps, ML Flow, CVAT, Amazon Web Services, Postman, JIRA, Voxel51